AP Chemistry: Stoichiometry **Bartonian ID _____**

Write neatly ◦ Boxes are reserved for *final answers only* ◦ *Bartonian ID* must be written on each page ◦ Do not staple.

1) How many moles of acetic acid, CH_3COOH, are contained within 24.71 g of acetic acid?

1

2) How many moles of sodium carbonate, Na_2CO_3, are contained by 57.3 g of sodium carbonate?

2

3) How many moles of sulfuric acid, H_2SO_4, are contained by 157 g of sulfuric acid?

3

4) How many grams of NaOH are contained within 0.785 moles of NaOH?

4

5) How many grams of C_6H_6 are present in 12.70 moles of C_6H_6?

5

6) How many grams of CCl_4 are present in 2.67 moles of CCl_4?

6

7) How many carbon atoms are contained in 84.3 g of ethyne (C_2H_2)?

7

8) How many ethyne molecules are contained in 84.3 g of ethyne (C_2H_2)?

8

9) C_2H_4 reacts with O_2 to form CO_2 and water.

a. Make a particulate drawing which shows that atoms are conserved during this reaction.

9a

b. Use your drawing to explain why all atoms are conserved during this reaction, and how this illustrates the law of conservation of mass.

9b

10) How many moles of hydrogen gas can be produced if 0.57 moles of hydrochloric acid reacts with excess solid zinc according to the following chemical equation?

$$2\ HCl + Zn \rightarrow H_2 + ZnCl_2$$

10

11) Nitrogen gas will react with hydrogen gas to produce ammonia. How many moles of hydrogen gas are required to produce 0.86 moles of NH_3?

11

12) N_2O_5 reacts with water to produce nitric acid. If 1.93 moles of N_2O_5 react with excess water, how many moles of nitric acid can be produced?

12

13) Suppose 1.65 moles of C_6H_6 react with excess oxygen to produce carbon dioxide and water.

a. How many moles of carbon dioxide will be produced in this reaction?

13a

b. How many moles of water will be produced in this reaction?

13b

c. How many moles of oxygen gas will be consumed during the reaction?

13c

14) How many grams of lithium are needed to produce 45.0 g of lithium nitride, according to the following process?

$$6\ Li(s) + N_2(g) \rightarrow 2\ Li_3N(s)$$

14

15) A 14.5 g sample of sodium chloride reacts with excess fluorine gas according to the following chemical equation.

$$2\ NaCl(s)\ +\ F_2(g)\ \rightarrow\ 2\ NaF(s)\ +\ Cl_2(g)$$

a. How many grams of sodium fluoride are produced?

15a

b. How many grams of chlorine gas are produced?

15b

16) What mass of P_2O_5 can be produced when a 172.1 g sample of phosphorus reacts with an excess of oxygen gas according to the following chemical equation?

$$4\ P\ +\ 5\ O_2\ \rightarrow\ 2\ P_2O_5$$

16

17) Determine the mass of lithium hydroxide that is produced when 12.87 g of lithium nitride reacts with an excess of water according to the following process.

$$Li_3N\ +\ 3\ H_2O\ \rightarrow\ 3\ LiOH\ +\ NH_3$$

17

18) Suppose 31.4 g of antimony (III) oxide reacts with excess carbon according to the following process.

$$Sb_2O_3 + 3\,C \rightarrow 2\,Sb + 3\,CO$$

a. What mass of antimony will be produced?

18a

b. What mass of CO will be produced?

18b

c. What mass of carbon is consumed during the reaction?

18c

19) A 24.5 g sample of sodium chloride reacts with 41.3 g of fluorine gas according to the following chemical equation.

$$2\ NaCl(s)\ +\ F_2(g)\ \rightarrow\ 2\ NaF(s)\ +\ Cl_2(g)$$

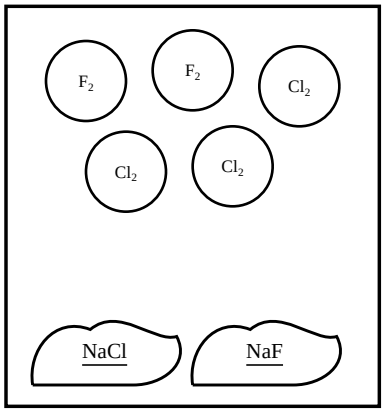
a. What is the limiting reactant? Justify your answer.

19a

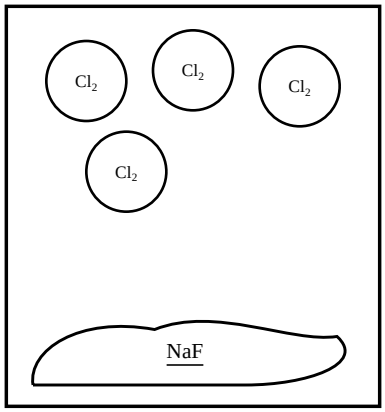
b. How many grams of chlorine gas are produced?

19b

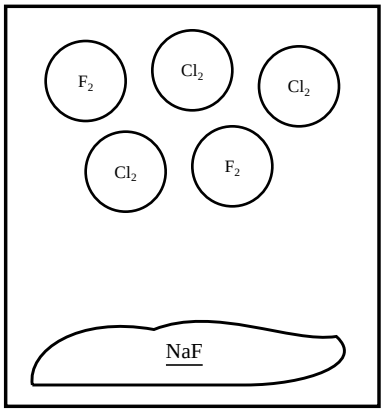
c. Which particulate representation below could be used to describe the species present in the reaction vessel after the process has gone to completion? Justify your answer.



(A)



(B)



(C)

19c

20) A 84.1 gram sample of phosphorus reacts with 85.0 g of oxygen gas according to the following chemical equation.

$$4\ P(s)\ +\ 5\ O_2(g)\ \rightarrow\ 2\ P_2O_5$$

a. Find the limiting reactant. Justify your answer.

20a

b. How many grams of P_2O_5 are produced in theory?

20b

c. If only 123 g of P_2O_5 are produced, what is the percentage yield?

20c

21) Most nitrogen exists in a gaseous state. Plants require a soluble form of nitrogen so they can absorb it from the ground. Ammonia is a good fertilizer, as the mass percent of nitrogen in ammonia is very high. The following reaction is used to convert gaseous nitrogen into ammonia, which can be used as fertilizer. Suppose 186.3 g of $N_2(g)$ react with 289.8 g of $H_2(g)$.

$$N_2(g) + 3\,H_2(g) \rightarrow 2\,NH_3$$

a. Which reactant is limiting? Justify your answer.

21a

b. What mass of ammonia can be produced in theory?

21b

c. If this reaction is known to have a 73.8% yield, what mass of ammonia could you expect to produce?

21c

22) A 5.75 g sample of silicon dioxide reacts with 5.50 g of sodium hydroxide according to the following chemical equation.

$$SiO_2(s) + 2\ NaOH(aq) \rightarrow Na_2SiO_3(aq) + H_2O(l)$$

a. What is the limiting reactant? Justify your answer.

22a

b. How many grams of $Na_2SiO_3(aq)$ are produced?

22b

c. What is the % yield if only 7.24 g of $Na_2SiO_3(aq)$ are produced?

22c

23) The reaction between phosphorus and liquid bromine is outlined below.

$$2\ P(s) + 3\ Br_2(l) \rightarrow 2\ PBr_3(l)$$

a. Identify the limiting reactant when 5.78 g of phosphorus react with 27.9 g of liquid bromine. Justify your answer.

23a

b. Based on your answer from part (a), determine the maximum mass of PBr_3 that can be produced in this reaction.

23b

c. If the actual yield of PBr_3 is found to be 22.3 g, find the percent yield in this reaction.

23c

24) The reaction between dihydrogen sulfide and sulfur dioxide is outlined below.

$$2\ H_2S(g)\ +\ SO_2(g)\ \rightarrow\ 3\ S(s)\ +\ 2\ H_2O(g)$$

a. Identify the limiting reactant when 3.89 g of dihydrogen sulfide react with 4.11 g of sulfur dioxide. Justify your answer.

24a

b. Based on your answer from part (a), determine the maximum mass of sulfur that can be produced in this reaction.

24b

c. If the actual yield of sulfur is found to be 4.89 g, find the percent yield in this reaction.

24c

25) A vitamin C tablet from a certain company contains 70.0 mg of ascorbic acid (Vitamin C). The entire tablet, however, has a mass of 0.825 g. Calculate the mass percent of vitamin C in the tablet.

25

26) A sample of sodium bicarbonate ($NaHCO_3$) is known to contain some impurities. It is found that Na makes up 18.00% of the entire mass of the sample. All of the Na comes from the $NaHCO_3$ compound. Find the mass percent of $NaHCO_3$ in the sample. (Hint: use ratios and cross multiply)

26

27) A sample of potassium chlorate ($KClO_3$) is known to contain some impurities. It is found that K^+ makes up 9.50% of the entire mass of the sample. All of the K^+ comes from the $KClO_3$ compound. Find the mass percent of $KClO_3$ in the sample.

27

28) Calculate the mass percent of each element in Na_2SO_4.

a. Mass percent of Na

28a

b. Mass percent of S

28b

c. Mass percent of O

28c

29) What is the empirical formula for C_4H_{10}?

29

30) What is the empirical formula for $C_2H_4O_2$?

30

31) Is CO_2 an empirical formula, a molecular formula, or both? Explain.

31

32) A hydrocarbon (a compound that contains only hydrogen and carbon) is found to be 74.5% carbon by mass.

a. Find the empirical formula for the compound.

32a

b. If the molar mass of the compound is 16.05 g/mol, find its molecular formula.

32b

33) It was found that a compound contained 68.1% carbon, 13.7% hydrogen, and 18.2% oxygen by mass.

a. Find the empirical formula for the compound.

33a

b. If the molar mass of the compound is 176.34 g/mol, what is its molecular formula?

33b

34) A sample of a hydrocarbon was analyzed and found to contain 12.00 g of carbon and 1.5 g of hydrogen.

a. Find the empirical formula of the compound. (Hint: as the masses are known, you do not need to assume you have a 100 g sample.)

34a

b. If the molar mass of the hydrocarbon is found to be 54.10 g/mol, what is its molecular formula?

34b

35) A sample of a compound was analyzed and found to contain 6.00 g of carbon and 1.100 g of hydrogen.

a. Find the empirical formula of the compound.

35a

b. If the molar mass of the compound is found to be 142.36 g/mol, what is its molecular formula?

35b

36) A 1.0857 gram sample of a compound containing only carbon, hydrogen, and oxygen was burned in excess oxygen gas. 3.190 g of carbon dioxide and 0.9360 g of water were produced. Find the empirical formula of the compound.

36

37) A 1.638 gram sample of a compound containing only carbon, hydrogen, and oxygen was burned in excess oxygen gas. 3.117 g of carbon dioxide and 1.911 g of water were produced. Find the empirical formula of the compound.

37